Neural Network Layer as Matrix Multiplication

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Setup: Neural Network Layer

$$Z = W^T X + B$$

$$\text{Input features} = 3, \qquad \text{Samples} = 2, \qquad \text{Neurons} = 2$$

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad W = \begin{bmatrix} 0.1 & 0.4 \\ 0.2 & 0.5 \\ 0.3 & 0.6 \end{bmatrix} \quad b = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix}$$

$$X \in \mathbb{R}^{3\times 2}, \qquad W \in \mathbb{R}^{3\times 2}, \qquad B \in \mathbb{R}^{2\times 2}$$

Neural Network Layer as Matrix Multiplication

$$Z = W^T X + B$$

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}, \qquad W = \begin{bmatrix} 0.1 & 0.4 \\ 0.2 & 0.5 \\ 0.3 & 0.6 \end{bmatrix}$$

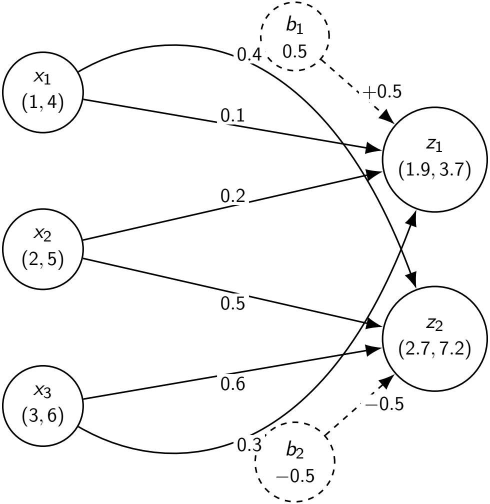
$$W^T = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.6 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix}$$

$$Z = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1.4 & 3.2 \\ 3.2 & 7.7 \end{bmatrix} + \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix} = \begin{bmatrix} 1.9 & 3.7 \\ 2.7 & 7.2 \end{bmatrix}$$

$$H = \sigma(Z)$$

Neural Network Layer with Connection Weights



$$Z = W^T X + B$$

Batch Normalization: Main Steps

Given a mini-batch activation matrix H, batch normalization follows four steps:

1. Compute the batch mean:

$$\mu_j = \frac{1}{m} \sum_{i=1}^{m} h_{ij}$$

2. Compute the batch variance:

$$\sigma_j^2 = \frac{1}{m} \sum_{i=1}^{m} (h_{ij} - \mu_j)^2$$

3. Normalize each activation:

$$\hat{h}_{ij} = \frac{h_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

4. Scale and shift:

$$y_{ij} = \gamma_j \hat{h}_{ij} + \beta_j$$